Lattices give us KEMs and FHE, but where are the efficient lattice PETs?

By Example of (Verifiable) Oblivious PRFs

Martin R. Albrecht
SandboxAQ

Outline

Oblivious PRFs

LWE and DH

The Trouble

Wrapping Up

Oblivious PRFs

Verifiable Oblivious PRFs

Client	Functionality	Server
$\xrightarrow{\quad m \quad}$		$\xleftarrow{\quad k \quad}$
	$c = F_k(m)$	
$\xleftarrow{\quad c \quad}$		$\xrightarrow{\quad \perp \quad}$

Example Applications: Privacy Pass

Privacy Pass

A privacy-enhancing protocol and browser extension.

Install:

- Home
- Protocol design
- FAQ
- Team
- Extension code
- Server code

Privacy Pass is a browser extension with the aim of making the internet more accessible.

Updates

Oct 2020: The Privacy Pass protocol is now in the process of being standardised by the IETF in the privacypass working group. All contributions are welcome! See the GitHub page for more details.

Oct 2019: Version 2.0 of the extension is now available in Chrome and Firefox!

April 2018: "Privacy Pass: Bypassing Internet Challenges Anonymously" accepted to PETS 2018. See below and in the FAQ for more details.

How?

Privacy Pass interacts with supporting websites to introduce an anonymous user-authentication mechanism. In particular, Privacy Pass is suitable for cases where a user is required to complete some proof-of-work (e.g. solving an internet challenge) to authenticate to a service. In short, the extension receives *blindly signed* 'passes' for each authentication and these passes can be used to bypass future challenge solutions using an *anonymous redemption* procedure. For example, Privacy Pass is supported by Cloudflare to enable users to redeem passes instead of having to solve CAPTCHAs to visit Cloudflare-protected websites.

DH-Based OPRF

Client		Server
	$c_x := H(x) \cdot g^r$ $\longrightarrow$	
	$\longleftarrow$ $d_x := c_x^k, c = g^k$	
$d_x / c^r = H(x)^k$		

Correctness: $d_x / c^r = c_x^k / c^r = (H(x) \cdot g^r)^k / (g^k)^r = H(x)^k$

LWE and DH

Lattices seem great

Bottom of the Stack:

- Kyber (KEM)
- Dilithium (Signature)
- Falcon (Signature)

Curve25519

Key generation	$\approx$ 60,000 cycles
Key agreement	$\approx$ 160,000 cycles
Public key	32 bytes
Key Share	32 bytes

https://eprint.iacr.org/2015/943

Top of the Stack:

- FHE [Gen09; BGV12; CGGI20; GSW13]
- iO [GGHRSW13; BDGM20]
- FE [SW05; BSW11]

Kyber-768

Key generation	$\approx$ 38,000 cycles
Encapsulation	$\approx$ 49,000 cycles
Decapsulation	$\approx$ 39,000 cycles
Ciphertext	1,088 bytes
Public key	1,184 bytes

https://bench.cr.yp.to/results-kem.html

The Learning with Errors Problem (LWE)

Given $(\mathbf{A}, \mathbf{c})$ with $\mathbf{c} \in \mathbb{Z}_q^m$, $\mathbf{A} \in \mathbb{Z}_q^{m \times n}$, $\mathbf{s} \in \mathbb{Z}_q^n$ and small $\mathbf{e} \in \mathbb{Z}^m$ is

$$\begin{pmatrix} \\ \\ \mathbf{c} \\ \\ \\ \end{pmatrix} = \begin{pmatrix} \leftarrow & n & \rightarrow \\ & & \\ & \mathbf{A} & \\ & & \\ & & \end{pmatrix} \times \begin{pmatrix} \\ \\ \mathbf{s} \\ \\ \\ \end{pmatrix} + \begin{pmatrix} \\ \\ \mathbf{e} \\ \\ \\ \end{pmatrix}$$

or $\mathbf{c} \leftarrow\$ \mathcal{U}\left(\mathbb{Z}_q^m\right)$.

LWE

$$\begin{pmatrix} c_0 \\ c_1 \\ c_2 \\ c_3 \\ c_4 \\ c_5 \\ c_6 \\ c_7 \end{pmatrix} = \begin{pmatrix} a_{0,0} & a_{0,1} & a_{0,2} & a_{0,3} & a_{0,4} & a_{0,5} & a_{0,6} & a_{0,7} \\ a_{1,0} & a_{1,1} & a_{1,2} & a_{1,3} & a_{1,4} & a_{1,5} & a_{1,6} & a_{1,7} \\ a_{2,0} & a_{2,1} & a_{2,2} & a_{2,3} & a_{2,4} & a_{2,5} & a_{2,6} & a_{2,7} \\ a_{3,0} & a_{3,1} & a_{3,2} & a_{3,3} & a_{3,4} & a_{3,5} & a_{3,6} & a_{3,7} \\ a_{4,0} & a_{4,1} & a_{4,2} & a_{4,3} & a_{4,4} & a_{4,5} & a_{4,6} & a_{4,7} \\ a_{5,0} & a_{5,1} & a_{5,2} & a_{5,3} & a_{5,4} & a_{5,5} & a_{5,6} & a_{5,7} \\ a_{6,0} & a_{6,1} & a_{6,2} & a_{6,3} & a_{6,4} & a_{6,5} & a_{6,6} & a_{6,7} \\ a_{7,0} & a_{7,1} & a_{7,2} & a_{7,3} & a_{7,4} & a_{7,5} & a_{7,6} & a_{7,7} \end{pmatrix} \cdot \begin{pmatrix} s_0 \\ s_1 \\ s_2 \\ s_3 \\ s_4 \\ s_5 \\ s_6 \\ s_7 \end{pmatrix} + \begin{pmatrix} e_0 \\ e_1 \\ e_2 \\ e_3 \\ e_4 \\ e_5 \\ e_6 \\ e_7 \end{pmatrix}$$

Performance

Storage: $\mathcal{O}(n^2)$; Computation $\mathcal{O}(n^2)$

Ring-LWE/Polynomial-LWE

$$\begin{pmatrix} c_0 \\ c_1 \\ c_2 \\ c_3 \\ c_4 \\ c_5 \\ c_6 \\ c_7 \end{pmatrix} = \begin{pmatrix} a_0 & -a_7 & -a_6 & -a_5 & -a_4 & -a_3 & -a_2 & -a_1 \\ a_1 & a_0 & -a_7 & -a_6 & -a_5 & -a_4 & -a_3 & -a_2 \\ a_2 & a_1 & a_0 & -a_7 & -a_6 & -a_5 & -a_4 & -a_3 \\ a_3 & a_2 & a_1 & a_0 & -a_7 & -a_6 & -a_5 & -a_4 \\ a_4 & a_3 & a_2 & a_1 & a_0 & -a_7 & -a_6 & -a_5 \\ a_5 & a_4 & a_3 & a_2 & a_1 & a_0 & -a_7 & -a_6 \\ a_6 & a_5 & a_4 & a_3 & a_2 & a_1 & a_0 & -a_7 \\ a_7 & a_6 & a_5 & a_4 & a_3 & a_2 & a_1 & a_0 \end{pmatrix} \cdot \begin{pmatrix} s_0 \\ s_1 \\ s_2 \\ s_3 \\ s_4 \\ s_5 \\ s_6 \\ s_7 \end{pmatrix} + \begin{pmatrix} e_0 \\ e_1 \\ e_2 \\ e_3 \\ e_4 \\ e_5 \\ e_6 \\ e_7 \end{pmatrix}$$

Performance (n is a power of two)

Storage: $\mathcal{O}(n)$; Computation $\mathcal{O}(n \log n)$

Ring-LWE/Polynomial-LWE

$$\sum_{j=0}^{n-1} c_j \cdot X^j = \left(\sum_{j=0}^{n-1} a_j \cdot X^j\right) \cdot \left(\sum_{j=0}^{n-1} s_j \cdot X^j\right) + \sum_{j=0}^{n-1} e_j \cdot X^j \bmod X^n + 1$$

$$c(X) = a(X) \cdot s(X) + e(X) \bmod \phi(X)$$

$$c_i = a_i \cdot s + e_i$$

We write $\mathcal{R} := \mathbb{Z}[X]/\phi(X)$ and $\mathcal{R}_q := \mathbb{Z}_q[X]/\phi(X)$.

Damien Stehlé, Ron Steinfeld, Keisuke Tanaka, and Keita Xagawa. **Efficient Public Key Encryption Based on Ideal Lattices.** In: *ASIACRYPT 2009*. Ed. by Mitsuru Matsui. Vol. 5912. LNCS. Springer, Berlin, Heidelberg, Dec. 2009, pp. 617–635. DOI: 10.1007/978-3-642-10366-7_36

Vadim Lyubashevsky, Chris Peikert, and Oded Regev. **On Ideal Lattices and Learning with Errors over Rings.** In: *EUROCRYPT 2010*. Ed. by Henri Gilbert. Vol. 6110. LNCS. Springer, Berlin, Heidelberg, 2010, pp. 1–23. DOI: 10.1007/978-3-642-13190-5_1

DH to Ring-LWE Dictionary

DH Land	Ring-LWE Land
g	a
g^x	$a \cdot s + e$
$g^x \cdot g^y = g^{x+y}$	$(a \cdot s + e_0) + (a \cdot t + e_1) = a \cdot (s + t) + e'$
$(g^a)^b = (g^b)^a$	$(a \cdot s + e) \cdot t = (a \cdot s \cdot t + e \cdot t)$ $\approx a \cdot s \cdot t \approx (a \cdot t + e) \cdot s$
(g, g^a, g^b, g^{ab}) $\approx_c (g, g^a, g^b, u)$	$(a,\ a \cdot s + e,\ a \cdot t + d,\ a \cdot s \cdot t + e')$ $\approx_c (a,\ a \cdot s + e,\ a \cdot t + d,\ u)$

Follow the Blueprint

C S

$$c_x := H(x) \cdot g^r \longrightarrow$$

$$\longleftarrow d_x := c_x^k, \quad c := g^k$$

$$d_x / c^r = H(x)^k$$

$$\begin{aligned} d_x / c^r &= c_x^k / c^r \\ &= (H(x) \cdot g^r)^k / (g^k)^r \\ &= H(x)^k \end{aligned}$$

C S

$$c_x := H(x) + a \cdot r + e \longrightarrow$$

$$\longleftarrow d_x := c_x \cdot k + e', \quad c := a \cdot k + e''$$

$$\lfloor 2/q \cdot d_x - c \cdot r \rceil$$

$$\begin{aligned} d_x - c \cdot r &= (H(x) + a \cdot r + e) \cdot k + e' - (a \cdot k + e'') \cdot r \\ &= H(x) \cdot k + a \cdot r \cdot k + e \cdot k + e' - a \cdot k \cdot r - e'' \cdot r \\ &= H(x) \cdot k + e \cdot k + e' - e'' \cdot r \\ &\approx H(x) \cdot k \end{aligned}$$

The Trouble

Trapdoor Friendly

The Problem

The server has to output $c_x \cdot k + e'$ for some $c_x \stackrel{?}{=} H(x) + a \cdot r + e$. This is not safe, the client can learn all the server's secrets with one maliciously crafted c_x.

- The client proves in zero-knowledge that c_x is well-formed: $c_x := H(x) + a \cdot r + e$
- This is expensive in terms of bandwidth and computation cost
  - [ADDS21]: $\approx$ 128GB per evaluation using [YAZXYW19]
  - [AG24]: $\approx$ 63kB per evaluation using [BS23]

Noise Leakage

The Problem

The client learns $e \cdot k + e' - e'' \cdot r$ where it chooses e and r.

- Drown everything with e': $\|e'\| \gg \|e \cdot k - e'' \cdot r\|$
- Since $q > \|e'\|$ we have that $q/\|e\|$ – the "signal to noise ratio" – is big.
- A big signal to noise ratio makes solving LWE easier.
- This requires us to use larger secret dimensions n to compensate.
- Bandwidth cost is essentially $n \log q$.

Noise Growth

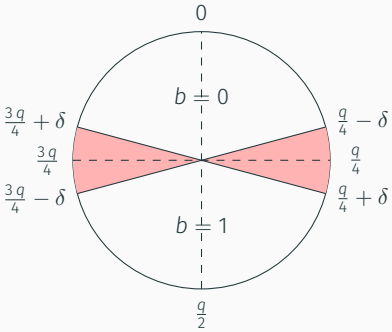
The Problem

Have:

$$H(x) \cdot k + e \cdot k + e' - e'' \cdot r \approx H(x) \cdot k.$$

Want:

$$\left\lceil \frac{2}{q} \cdot \left(H(x) \cdot k + e \cdot k + e' - e'' \cdot r \right) \right\rceil = \left\lceil \frac{2}{q} \cdot \left(H(x) \cdot k \right) \right\rceil$$



- Make $q \gg \text{poly}(\lambda)$ so that red area is negligibly small.
- Big "signal-to-noise" ratio: increasing n again.

Wrapping Up

Realisations of this Blueprint

Work	Model	1-time Offline	Online	Q
[ADDS21]	H-H	–	2MB	
[ADDS21]	M-M	–	128GB	
[AG24]	M-M	114kB	198kB	2^{32}
[ESTX24]	M-M	20kB	159kB	2^{32}

H: semi-honest, M: malicious

- Martin R. Albrecht, Alex Davidson, Amit Deo, and Nigel P. Smart. **Round-Optimal Verifiable Oblivious Pseudorandom Functions from Ideal Lattices.** In: *PKC 2021, Part II*. ed. by Juan Garay. Vol. 12711. LNCS. Springer, Cham, May 2021, pp. 261–289. DOI: 10.1007/978-3-030-75248-4_10
- Martin R. Albrecht and Kamil Doruk Gür. **Verifiable Oblivious Pseudorandom Functions from Lattices: Practical-Ish and Thresholdisable.** In: *ASIACRYPT 2024, Part IV*. ed. by Kai-Min Chung and Yu Sasaki. Vol. 15487. LNCS. Springer, Singapore, Dec. 2024, pp. 205–237. DOI: 10.1007/978-981-96-0894-2_7
- Muhammed F. Esgin, Ron Steinfeld, Erkan Tairi, and Jie Xu. **LeOPaRd: Towards Practical Post-Quantum Oblivious PRFs via Interactive Lattice Problems.** Cryptology ePrint Archive, Report 2024/1615. 2024. URL: https://eprint.iacr.org/2024/1615

An Alternative from FHE

We do have efficient FHE: ciphertexts are typically **smaller** than messages above.

- **Simple idea:**
  1. Client FHE encrypts x as $[x]$
  2. Server homomorphically computes PRF using plaintext k and $[x]$ to obtain $[F_k(x)]$
  3. Client FHE decrypts $F_k(x)$
- **Problem:** PRFs need deep circuits, deep circuits are expensive
- **Proposal:** Use Dark Matter (weak-)PRF candidate [BIPSW18] $\sum(\mathbf{A} \cdot \mathbf{x} \bmod 2) \bmod 3$ where $\mathbf{A}$ is the secret key

Martin R. Albrecht, Alex Davidson, Amit Deo, and Daniel Gardham. **Crypto Dark Matter on the Torus - Oblivious PRFs from Shallow PRFs and TFHE**. In: *EUROCRYPT 2024, Part VI*. ed. by Marc Joye and Gregor Leander. Vol. 14656. LNCS. Springer, Cham, May 2024, pp. 447–476. DOI: 10.1007/978-3-031-58751-1_16

Overview of Round-Optimal OPRFs w/o Trusted Setup

Work	Assumption	Model	1-time Offline	Online
ADDS21	(R)LWE+SIS	H-H	-	2MB
ADDS21	(R)LWE+SIS	M-M	-	128GB
AG24	(R)LWE+SIS	M-M	114kB	198kB
ADDG24	mod(2,3)+lattices	M-H	2.5MB	10KB
ADDG24	mod(2,3)+lattices	M-M	2.5MB	160KB
ESTX24	iMLWER-RU+MLE+SIS	M-M	20kB	159KB
APRR24	mod(2,3)	M-H, pp	4.75B	114.5B
FOO23	AES+Garbled Circuits	H-H	-	6.79MB
Basso24	Higher-Dimensional isogenies	M-M	-	28.9kB
HHM+23	Isogenies F_p + lattices + HE OT	H-H	-	640kB
dSP23	Isogenies F_p	M-H, pp	68.4 kB	384B
dSP23	Isogenies F_p	M-H, pp	-	16.38kB

adapted from `https://heimberger.xyz/oprfs.html`

Thank You

https://ia.cr/2019/1271

https://ia.cr/2023/232

https://ia.cr/2024/1459

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